Southern New Hampshire University

CS 255 Business Requirements Document Template

System Components and Design

Purpose

- What is the purpose of this project? Who is the client and what do they want their system to be able to do?
- **Client Overview:** The client is DriverPass, owned by Liam, seeking to capture a market void in preparing students for local Department of Motor Vehicles (DMV) driving exams.
- **Core Purpose:** To build a flexible web-based system that manages online driver training classes, practice exams, and behind-the-wheel driving lesson reservations.
- **Primary Scope:** Deliver a cloud-hosted platform allowing students, drivers/instructors, secretaries, IT officers, and management to access data online from any desktop or mobile device.

System Background

- What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?
- **Problem Statement:** Many students fail their DMV driving tests due to inadequate preparation. DriverPass aims to address this by offering structured online practice exams combined with practical behind-the-wheel training.
- **Core Business Offerings (Packages):**
- **Package One:** 6 hours of behind-the-wheel training with a driver (spread over three 2-hour sessions).
- **Package Two:** 8 hours of behind-the-wheel training plus an in-person lesson on DMV rules and policies.
- **Package Three:** 12 hours of behind-the-wheel training, an in-person DMV rules lesson, and full access to online classes with study content and practice tests.
- **Essential System Components:**
- **Online Scheduling & Reservation Engine:** Enables students or the company secretary to book, cancel, or modify 2-hour driving appointments.
- **Resource Matching & Fleet Tracker:** Matches specific students to specific drivers, time slots, and cars (across DriverPass's fleet of 10 cars).
- **Online Testing & Progress Portal:** Displays practice test statuses (Not Taken, In Progress, Failed, Passed), completion times, and test scores.
- **Driver Notes & Student Dashboard:** Displays driver lesson logs, lesson start/end hours, pickup/drop-off locations, driver comments, special needs, and photos.
- **DMV Compliance Module:** Connects to the DMV to receive automated notifications and updates regarding new rules, policies, and sample test questions.

Objectives and Goals

- What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?
- **Goal 1 (Cloud & Remote Access):** Enable online data access from any computer or mobile device while restricting data updates/modifications to online modes to prevent server data redundancy.
- **Goal 2 (Appointment Scheduling):** Allow online self-service booking, modification, and cancellation of 2-hour driving lessons, as well as phone/in-person booking by the secretary.
- **Goal 3 (Audit Logging & Accountability):** Log all record creations, modifications, and cancellations with user attribution to generate downloadable activity reports.
- **Goal 4 (Reporting & Exporting):** Allow management to export reports and operational information into local tools like Microsoft Excel for home use.
- **Goal 5 (DMV Syncing):** Receive automated update notifications whenever the DMV updates policies or sample exam questions.

Requirements

Nonfunctional Requirements

Performance Requirements

- What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?
- The system shall run as a cloud-hosted web application accessible via web browsers on desktop computers and mobile devices.
- Data modifications and updates shall only be permitted while connected online to prevent data duplication and redundancy across servers.
- System maintenance, cloud security updates, and automated backups shall be managed transparently in the cloud environment to ensure minimal operational downtime.

Platform Constraints

- What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?
- The web interface shall be cross-platform compatible across any standard modern web browser on Windows, macOS, iOS, and Android.

- The backend shall be cloud-hosted to offload local hardware, backup, and infrastructure maintenance.
- The backend shall utilize a relational database to maintain persistent records for students, drivers, 10 fleet vehicles, schedules, packages, and activity logs.
- Data must be formatted to allow users to download and work on reports using local spreadsheet applications such as Microsoft Excel.

Accuracy and Precision

- How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?
- The system shall distinguish between user types via role-based account permissions separating Liam (Owner), Ian (IT Officer), Secretary, Drivers, and Students.
- Practice exam tracking must accurately record and display the exact test name, time taken, numeric score, and exact status (Not Taken, In Progress, Failed, or Passed).
- Driving lesson logs must precisely record lesson start hour, end hour, assigned driver, assigned car, pickup location, drop-off location, and driver comments.
- The system shall log every schedule of creation, modification, and cancellation, identifying the responsible user for audit reports.

Adaptability

- Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?
- The system shall allow IT Officer Ian to manage user accounts, reset lost passwords, and block access for terminated employees without code changes.
- The system shall allow management to disable specific training packages, so new customers cannot register for them.
- The system architecture shall support external connections to DMV systems to receive push notifications when new rules, policies, or sample questions are released.

Security

- What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a “brute force” hacking attempt? What happens if the user forgets their password?
- **User Authentication:** Access shall be controlled via role-based user accounts and passwords.

- **Password Self-Service:** The system shall provide an automated self-service feature allowing customers to reset forgotten passwords over the internet.
- **Administrative Control:** IT Officer Ian shall possess override authority to reset passwords or block user accounts immediately.
- **Data Security & Cloud Infrastructure:** Security management, data encryption in transit/rest, and cloud backups shall be handled at the cloud infrastructure level to minimize technical burden on DriverPass.
- **Payment Security:** Customer registration data including first name, last name, address, phone number, state, credit card number, expiration date, security code, pickup location, and drop-off location must be securely encrypted during transmission.

Functional Requirements

- The system shall allow new customers to register by capturing first name, last name, address, phone number, state, pickup location, drop-off location, credit card number, expiration date, and security code.
- The system shall enforce that the customer's drop-off location defaults to the pickup location.
- The system shall provide automated password reset functionality for users via the internet.
- The system shall allow IT Officer Ian to manage user accounts, reset user passwords, and block access for terminated employees.
- The system shall allow students to select and register for one of three training packages (Package One, Package Two, or Package Three).
- The system shall allow management to disable training packages to prevent new student registrations.
- The system shall allow students and the secretary to schedule, modify, or cancel 2-hour driving lesson reservations online.
- The system shall track and assign a driver, a specific car (from the 10-car fleet), time slot, student, pickup location, and drop-off location for every driving lesson.
- The system shall track all reservation creations, modifications, and cancellations, capturing the user ID and timestamp for audit activity reports.
- The system shall display online practice test progress, including test name, time taken, score, and status (Not Taken, In Progress, Failed, Passed).
- The system shall display driver notes and lesson logs, including lesson start hour, end hour, and driver comments.
- The system shall display student photo, driver photo, and special needs information on the student dashboard layout.
- The system shall provide a "Contact Us" page and a mechanism for DriverPass to contact the student.
- The system shall receive automated notifications from the DMV when new rules, policies, or sample questions are published.

- The system shall allow management to download operational reports and information in formats compatible with Microsoft Excel.

User Interface

- What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?
- **General UI Principles:** Web-based, responsive cloud interface designed according to the layout sketch provided by Liam.
- **Interface Users & Needs:**
- **Student UI:** Web interface displaying Logo at the top, Online Test Progress module, Student Information panel, Driver Notes table, Special Needs block, Driver Photo, Student Photo, and booking controls.
- **Secretary UI:** Input forms to enter student registration details, take phone calls, and schedule, modify, or cancel driving appointments.
- **Driver UI:** Input view to leave comments and log start/end hours for each 2-hour lesson.
- **IT Admin UI (Ian):** User account controls, access blocking tools, password reset overrides, and system activity log viewer.
- **Owner UI (Liam):** Activity report generator, package disabler toggle, and downloadable report export controls.
- **Interaction Methods:** Accessed via web browsers across desktop computers, tablets, and mobile devices.

Assumptions

- What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?
- **Internet Connection:** It is assumed users have active internet access when performing data updates or reservations, as offline editing is restricted to prevent server data redundancy.
- **Third-Party Payment Processing:** It is assumed that credit card processing is handled via a secure payment gateway integrated into the cloud platform.
- **DMV API Availability:** It is assumed the local DMV provides an accessible digital interface or subscription feed to push updates and sample questions.
- **Fleet Availability:** It is assumed all 10 vehicles are equipped for student driver instruction and assigned to designated drivers.

Limitations

- Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?
- **No Offline Data Modifications:** Users cannot modify or update system data while offline to prevent data duplication and redundancy across servers.
- **Fixed Module Customization in Phase 1:** Customizing or creating entirely new package modules requires software developer intervention and cannot be performed by non-developers in the initial release.
- **Schedule Timeline:** Project completion is bound by the project schedule outlined by Sam and Jennifer (January 22 to May 10).
- **Web-Only Scope:** Native mobile applications (iOS/Android apps) are excluded from Phase 1; access is web-browser based.

Gantt Chart

